Eating-Disorder References in Online Suicidal Distress Narratives: An LLM-Assisted Analysis of Mental Health Discourse

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Abstract

Eating disorders (EDs) are frequently associated with suicidal distress (SD), yet less is known about how ED-related experiences are disclosed in online suicidal-distress communities. This cross-sectional study examines post-level ED references, demographic self-disclosures, co-occurring mental health condition references, and substance-use references in Reddit's *r/SuicideWatch* community. Using an LLM-assisted keyword-based filtering framework followed by human validation, we identified 688 ED-related posts within a corpus of 27,434 SD posts, corresponding to a post-level co-occurrence rate of 2.5% (95% CI: 2.33%–2.70%). Among posts with self-reported gender (n = 75, 10.9% of the ED-related corpus), female-authored posts were most represented (42.7%). Among posts with identifiable age (n = 170, 24.7%), posts from individuals aged 18–20 were most represented (28.0%). These demographic findings are exploratory because they are based on small, self-disclosed subsamples. Co-occurring mental health conditions were referenced in 79.0% of ED-related posts (95% CI: 75.8%–81.9%), particularly depression (43.9%) and anxiety (31.5%). Substance use was referenced in 27.0% of posts, with alcohol-related terms appearing most frequently. Overall, this study provides a corpus-level characterization of explicit ED–SD co-disclosure in online mental health discourse and highlights methodological considerations for NLP research using self-disclosed, post-level mental health data.

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1 Introduction

Eating disorders (EDs) are severe psychiatric conditions characterized by persistent disturbances in eating behavior, distorted body image, and maladaptive weight-control practices [1, 2]. EDs are also associated with elevated mortality, with suicide representing an important contributor to premature death [3]. Clinical and longitudinal studies consistently document elevated suicide risk among individuals with EDs [4], with risk further complicated by comorbid depression, personality disorders, and substance misuse [5, 6].

Despite this clinical literature, comparatively less is known about how ED-related experiences are disclosed in online communities where individuals discuss suicidal distress. Clinical and survey-based studies provide standardized evidence about ED–SD associations, but they may not fully capture how people describe these experiences in everyday, self-authored discourse [7]. Social media platforms such as Reddit provide one complementary source of naturalistic mental health language, particularly in anonymous or pseudonymous communities where users may disclose distress, barriers to care, and co-occurring psychosocial concerns [7, 8, 23, 26, 27].

Within this context, *r/SuicideWatch* contains first-person narratives related to suicidal distress. These posts should not be treated as clinical records or diagnostic evidence, but they can support post-level corpus analysis of how ED-related language, psychiatric comorbidity references, and substance-use references appear in suicidal-distress discourse. This framing is important for responsible NLP research: the goal is to characterize explicit self-disclosed language patterns, not to infer individual diagnoses or clinical risk.

Prior research on EDs in social media contexts has largely focused on exposure effects on body image disturbance and symptom development [9, 10], participation in pro-ED communities [11], or automated detection and characterization of ED-related language [12]. However, systematic corpus-level characterization of ED references specifically within suicidal-distress posts remains limited, particularly with respect to demographic self-disclosures, co-occurring mental health condition references, and substance-use references.

This study addresses that gap with four research questions:

- **RQ1:** What is the post-level rate of ED references in Reddit's *r/SuicideWatch* forum, and which ED indicators are most frequently mentioned?
- **RQ2:** What age and gender patterns are observed in the subset of posts containing self-disclosed demographic information?
- **RQ3:** Which mental health conditions most commonly co-occur with ED references in these posts?
- **RQ4:** What patterns of substance use are observed in these posts?

We analyze 27,434 posts from Reddit's *r/SuicideWatch* subreddit. Using a keyword-based framework augmented with LLM-assisted keyword generation—chosen for interpretability, reproducibility,

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and suitability for corpus construction—we identify 688 posts explicitly referencing EDs, corresponding to a 2.5% post-level co-occurrence rate (95% CI: 2.33%–2.70%). Demographic attributes are extracted only from posts containing explicit self-disclosures, and these findings are interpreted as exploratory rather than population-level estimates. We also examine co-occurring mental health condition references and substance-use references within the ED-related subset.

Contributions: This study makes three primary contributions:

(1) A validated corpus of 688 ED-related posts from *r/SuicideWatch*, constructed using LLM-assisted keyword expansion followed by manual review with substantial inter-annotator agreement.
(2) A systematic post-level characterization of explicit ED–SD co-disclosure, including the most frequently observed ED-related indicators.
(3) A descriptive analysis of post-level psychosocial attributes within ED-related posts, including demographic self-disclosures, mental health condition references, and substance-use references, with explicit attention to the limits of post-level online mental health inference.

2 Related Work

2.1 Eating Disorders and Suicidal Risk

Clinical research consistently documents elevated suicide risk among individuals with eating disorders [13, 14]. Pisetsky et al. [4] reported higher suicide attempt rates among women with EDs, with purging behaviors strongly associated with elevated risk. Cliffe et al. [5] found that 6.7% of 4,895 ED patients attempted suicide over a 10-year period, with anorexia nervosa carrying the highest risk and comorbid personality disorder, depression, and substance misuse further elevating it. Bulik et al. [6] reported that 16.9% of individuals with anorexia nervosa had attempted suicide. Brown et al. [15] demonstrated that even sub-threshold binge eating is associated with suicidality. Smith et al. [16] identified emotion dysregulation, trauma, and low body regard as shared contributors to both EDs and suicidality. Milos et al. [17] similarly found that 26% of women with EDs had attempted suicide, with rates highest among those with purging subtypes and affective or personality disorder comorbidities.

2.2 Eating Disorders in the Context of Social Media

Research on EDs in social media contexts has followed two primary directions: the effects of social media on ED development, and the computational detection of ED-related content. Dane and Bhatia [9] found that social media use is associated with increased body image concerns via social comparison. Dondzilo et al. [10] showed that temporarily reducing social media use led to a significant decrease in ED symptoms. On the computational side, Sukunesan et al. [11] examined pro-ED communities on Twitter and found harmful content perpetuated by a small subset of highly active users. Chu et al. [12] introduced LLM-based frameworks to characterize online ED communities and detect harmful content. Both studies primarily focus on identifying and moderating harmful ED content, rather than characterizing the mental health profiles of individuals already in suicidal crisis.

2.3 Research Gap

A clear gap emerges: clinical research has thoroughly documented the association between EDs and suicidal risk, while social media studies have examined ED-related content in online environments. However, less work has systematically characterized explicit ED-related language within suicidal-distress posts and examined the accompanying post-level context, including demographic self-disclosures, co-occurring mental health condition references, and substance-use references. This study addresses that gap through a validated, corpus-level analysis of ED-related posts in *r/SuicideWatch*.

3 Dataset Description and Statistics

3.1 Data Collection

The textual data analyzed in this study represent posts from the Reddit community *r/SuicideWatch*. These posts were drawn from a publicly available Kaggle dataset, which we further processed in our earlier work, resulting in 27,434 posts[1]. The dataset includes only posts in which users explicitly disclose personal experiences related to suicidal thoughts, excluding comments and responses, and covers the period from December 16, 2008, to January 2, 2021.

3.2 Keyword Selection and Corpus Construction

To identify ED-related posts, we applied a keyword-based filtering strategy to the corpus. We deliberately chose this approach—augmented with LLM-assisted expansion of candidate terms—over direct LLM classification for three reasons: (1) *interpretability*, since the keyword list is fully auditable and human-reviewable; (2) *reproducibility*, as regex-based keyword matching produces deterministic outputs independent of model versioning or API access; and (3) *suitability for corpus construction*, because a transparent filtering mechanism facilitates principled manual validation and inter-annotator agreement assessment. Although this design sacrifices some contextual sensitivity compared to zero- or few-shot LLM classification, it provides a reliable and inspectable foundation for corpus-level analysis.

Initial keywords were derived from established psychiatric diagnostic definitions and prior NLP studies on ED-related social media content. The keyword list was subsequently expanded using two large language models—GPT-4 (OpenAI, March 2023) and LLaMA-2 (Meta, 2023)—to capture broader clinical, behavioral, and community-specific terminology. Using two complementary models helps reduce model-specific lexical bias and improve coverage. The combined outputs were systematically reviewed to remove redundant, ambiguous, or non-self-referential terms.

Keywords cover three primary domains: (1) clinical terminology (e.g., "anorexia", "bulimia", "binge-eating", "ARFID", "EDNOS"); (2) behavioral expressions (e.g., "purging", "calorie restriction", "excessive exercise"); and (3) psychological traits (e.g., "body shame", "body dissatisfaction"). Using this keyword set, 688 ED-related posts were identified, representing 2.5% of the full dataset.

[1] https://github.com/sazzadcsedu/SocialDemopExtract.git

3.3 Corpus Validation

To validate the corpus, 150 randomly sampled posts were independently annotated by two trained reviewers using pre-defined guidelines. Posts were considered ED-relevant if they contained self-referential descriptions of disordered eating behaviors, ED-related symptoms, or associated psychological experiences, while non-qualifying mentions (informational, metaphorical, or third-person references) were excluded. Inter-annotator agreement was substantial (88.67% raw agreement; Cohen's $\kappa = 0.67$). Disagreements were resolved through discussion to generate consensus gold-standard labels. Automated keyword-based labeling was then evaluated against this gold standard, achieving 82% agreement.

It is important to note that this validation assesses *precision* only, confirming that posts identified by keywords are genuinely ED-related. Because the full corpus was not independently annotated, *recall* cannot be estimated; some posts using implicit, metaphorical, or evolving ED-related language may have been missed. The reported 2.5% post-level co-occurrence rate should therefore be interpreted as a lower-bound estimate. This limitation is discussed further in Section 7.

3.4 Corpus Statistics

The final corpus ($n = 688$) has a mean post length of 607.65 words (SD = 477.06), with individual posts ranging from 161 to 3,509 words. The most frequently occurring keywords include *eating disorder* (267 posts), *anorexia* (117), *calories* (62), *binge eating* (56), *bulimia* (48), *body image* (47), *body dysmorphia* (46), and *weight loss* (44). These keywords reflect a wide range of issues mentioned in the posts, encompassing both clinical conditions (e.g., *anorexia*, *bulimia*) and associated psychological and physical symptoms (e.g., *body image concerns*, *weight loss*).

4 Methods

4.1 Demographic Extraction

We use the demographic information extraction tool SocialDemoExtract [18] to identify self-disclosed age and gender attributes. The tool assigns gender labels to one of three categories: male, female, and gender minority (encompassing non-binary, genderqueer, and other self-identified gender-diverse expressions).

Because self-reported demographic attributes are typically absent from online posts, demographic information was identifiable in only a small subset: gender in 75 posts (10.9% of the corpus) and age in 170 posts (24.7% of the corpus). Accordingly, all demographic findings are interpreted as *exploratory and preliminary* rather than representative population-level estimates. Reported percentages reflect proportions within these subsamples, and should be interpreted with caution when drawing broader inferences.

4.2 Co-occurring Mental Health Conditions

To identify co-occurring mental health conditions, candidate keywords were generated using GPT-4o and Gemini 2.5 Flash, each prompted to produce a comprehensive list of mental health conditions and related terms commonly referenced in SD-related social media discussions. The two outputs were merged into a unified candidate list and subsequently refined through a structured manual review process by the research team. This refinement involved removing terms that were overly broad, clinically imprecise, or unlikely to appear in self-referential social media discourse, as well as adding clinically relevant terms not captured by either LLM. The final list comprises 80 terms spanning mood disorders, anxiety disorders, trauma-related conditions, personality conditions, neurodevelopmental conditions, and behavioral concerns.

To validate the keyword-based approach for identifying mental health conditions in posts, a random subset of 125 posts was manually reviewed to determine whether each accurately reflected the author's mental health condition. Posts in which the keyword did not correspond to the author's mental health condition were labeled as negative.

4.3 Co-occurring Substance Use

Substance use keywords were developed from two complementary sources: the clinical literature on ED-substance use comorbidity [19], and manual review of a randomly sampled subset of 100 corpus posts to capture the informal register used in online discourse. Keywords were organized into seven pharmacological categories: alcohol, cannabis, opioids, stimulants, prescription medications, psychedelics, and other substances. Each category includes both clinical and colloquial terms to capture variation in online discourse (e.g., alcohol: drink, booze; cannabis: weed, THC; opioids: heroin, painkillers; stimulants: cocaine, speed; prescription medications: Xanax, benzos; psychedelics: LSD, shrooms; other substances: MDMA, molly).

5 Results and Discussion

5.1 Post-Level Rate of ED References (RQ1)

Of 27,434 SD posts, 688 contained ED-related content, yielding a post-level co-occurrence rate of 2.5% (95% CI: 2.33%–2.70%)—approximately 1 in 40 posts within this already high-risk community. The most frequent keywords were *eating disorder* (267 posts, 38.8%), *anorexia* (117, 17.0%), *calories* (62, 9.0%), *binge eating* (56, 8.1%), *bulimia* (48, 7.0%), *body image* (47, 6.8%), *body dysmorphia* (46, 6.7%), and *weight loss* (44, 6.4%).

The dominance of the generic term *eating disorder* over specific diagnostic labels suggests that posts frequently employ broad self-identification language rather than referencing a formally specified diagnosis. For NLP-based risk detection, this has methodological implications: systems anchored solely to diagnostic terminology may fail to capture a substantial portion of ED-related suicidal discourse. The prominence of behavioral and perceptual terms (*calories*, *body image*, *body dysmorphia*) alongside clinical labels underscores the variability in how ED-related distress is expressed.

These linguistic patterns are illustrated in post excerpts. One post states: *"I've lived longer with an eating disorder than without one (it probably started when I was 8–9) and don't know what I'd do without that feeling of control,"* highlighting the ego-syntonic nature of disordered eating, which can complicate both clinical intervention and automated risk detection. Another post illustrates how perceived barriers to care may redirect disclosure toward online communities: *"I have sadly started to eat less/control how much I eat,*

which is eating back to my anorexia. When I tried to get help…I was rejected because I was at a healthy weight."

5.2 Demographic Patterns in Self-Disclosing Posts (RQ2)

The demographic analysis revealed trends in age and gender among individuals discussing eating disorders alongside suicidal ideation. As these findings are based on a small subset of posts containing self-reported demographic information, they should be interpreted as exploratory.

Age: Among the 170 posts with identifiable age (24.7% of the corpus), the largest groups, listed in order of representation, were 18–20, 15–17, and 21–24 years, with the 18–20 group most represented (28.0%, 95% CI: 24.8%–31.5%), closely followed by the 15–17 and 21–24 groups. This distribution is consistent with clinical data indicating peak ED onset between ages 15 and 19 [20]. A high representation from the 21–24 age group (≈23%) indicates that ED-related suicidal crisis expressions extend into early adulthood. Below are sample excerpts representing different age groups:

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*"…right now, I'm **16 years old**, trying to live a normal life in the suburbs. There's no way I can deal with this disease anymore…"*

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*"…I'm almost **18 years old** and have been suffering from depression and some kind of eating disorder for about 7 years already…"*

Gender: Among the 75 posts with self-reported gender (10.9% of the corpus), 42.7% (95% CI: 39.1%–46.4%) were authored by females, 33.3% by males, and 24.0% by gender-minority individuals. A sample excerpt illustrating female representation reads:

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*"I am a 20-year-old **female**… I think about food constantly. I love food, especially that unhealthy, greasy, salty nastiness that everyone says you should cut out of your life forever…"*

The presence of female, male, and gender-minority self-disclosures is consistent with clinical literature showing that ED-related distress is not limited to one demographic group [2, 21]. However, because gender was identifiable in only 75 posts, these results should be interpreted cautiously as patterns within the self-disclosing subset rather than as representative estimates of the broader ED-related corpus.

5.3 Co-occurring Mental Health Conditions (RQ3)

Mental health conditions were referenced in 79.0% (95% CI: 75.8%–81.9%) of ED-related SD posts—four out of five posts referencing at least one co-occurring condition. This high rate of co-referenced conditions demonstrates that ED-related SD posts are rarely clinically circumscribed but are instead embedded within a dense network of intersecting psychological conditions. Table 1 summarizes the most commonly referenced conditions.

Depression (43.9%) and anxiety (31.5%) predominated, consistent with clinical evidence linking emotional dysregulation and negative affectivity to ED pathology [22]. Trauma and PTSD (22.1%) reflect well-documented associations between adverse life events and both ED onset and suicidal behavior. Guilt and shame (18.6%) capture the self-critical cognitive style that sustains both disorder maintenance and suicidal distress through cycles of restriction, transgression, and self-punishment.

Table 1: Prevalence of Co-occurring Mental Health Conditions in ED-Related SD Posts (n=688)

Condition	Prevalence (%)	95% CI
Depression	43.9	40.2–47.6
Anxiety	31.5	28.2–35.1
Trauma / PTSD	22.1	19.2–25.4
Guilt / Shame	18.6	15.9–21.7
OCD	10.3	8.2–12.9
Self-harm	9.7	7.6–12.2
Borderline PD	7.8	6.0–10.2
Autism Spectrum	2.3	1.4–3.9

Posts could mention multiple conditions; percentages are not mutually exclusive and sum to >100%.

OCD (10.3%), borderline personality disorder (7.8%), and self-harm (9.7%) collectively identify a subset characterized by severe affect dysregulation and impulsive self-destructive behavior, consistent with clinical patterns reported in ED populations with purging subtypes [24]. ASD-related keywords appeared in 2.3% of posts.

It should be noted that without a comparative baseline of non-ED SD posts, it is not possible to determine whether these comorbidity rates are uniquely elevated at the ED–SD intersection or reflect patterns common to suicidal distress posts more broadly. This is an important analytical constraint, discussed further in Section 7.

The manual review of 125 randomly selected posts confirmed that 87.2% (n = 109) accurately reflected the corresponding mental health condition, supporting the reliability of the keyword-based method for capturing self-reported co-occurring conditions.

Overall, these results show that ED-related suicidal-distress posts often include multiple co-occurring mental health references. This supports the value of corpus-level analysis for describing how overlapping distress signals appear in self-disclosed online mental health discourse.

5.4 Co-occurring Substance Use (RQ4)

Substance use was referenced in 27.0% of posts (n = 186; 95% CI: 23.7%–30.6%). Table 2 summarizes the most frequent substance use terms. Alcohol predominated, with *drinking*, *drink*, and *alcohol* collectively referenced in 17.7% of posts, consistent with prior literature identifying alcohol use as a maladaptive coping strategy for emotional dysregulation in ED populations [19].

Cannabis (4.2%) was also referenced, potentially reflecting attempts to manage anxiety or appetite-related distress. Mentions of benzodiazepines alongside SD may signal acute distress, consistent with clinical observations that these medications are involved in overdose mortality among individuals with psychiatric comorbidity. Illicit substances—including heroin (1.2%), methamphetamine (0.7%), and fentanyl (0.4%)—align with clinical observations of multi-impulsivity in certain ED subgroups [25].

Table 2: Most Frequent Substance Use Terms in ED-Related SD Posts (n=688)

Term	Category	#Mentions	Mentions (%)
Drinking	Alcohol	47	6.8
Drink	Alcohol	38	5.5
Alcohol	Alcohol	37	5.4
Weed/Cannabis	Cannabis	29	4.2
Sleeping pills	Prescription	11	1.6
Xanax	Prescription	10	1.5
Speed	Stimulant	9	1.3
Heroin	Opioid	8	1.2
Meth	Stimulant	5	0.7

Manual review of a subset of posts confirmed that approximately 88% (109 out of 125) of keyword-identified mentions accurately reflected the poster's substance use, supporting the reliability of these estimates.

Overall, the patterns observed are consistent with clinical reports that substance-use concerns can co-occur with eating-disorder symptoms [25]. In this corpus, substance-use references should be interpreted as post-level textual indicators rather than evidence of clinical substance-use disorder.

6 Implications for NLP Analysis of Eating Disorder and Suicidal Distress Discourse

This study has three implications for NLP research on online mental health discourse.

Transparent corpus construction: The corpus was constructed using LLM-assisted keyword expansion followed by manual review and validation. This design is useful for corpus-level studies because the final keyword set is inspectable and the retrieval process is reproducible. At the same time, this approach should be interpreted as high-precision retrieval of explicit ED-related references rather than exhaustive identification of all ED-related distress in the source corpus.

Modeling overlapping post-level signals: Mental health condition references appeared in 79.0% of ED-related posts, and substance-use references appeared in 27.0%. These findings indicate that ED-related suicidal-distress posts often contain multiple overlapping psychosocial signals. NLP approaches that represent only one dominant label per post may therefore miss important contextual information. Future work could examine multi-label or span-level approaches that distinguish ED-related language from co-occurring mental health and substance-use references.

Careful interpretation of self-disclosed demographics: Age and gender were identifiable only in subsets of posts. Therefore, demographic results are useful for describing the self-disclosing subset but should not be generalized to all users or to clinical populations. NLP studies using similar online data should report the amount of missing demographic information and avoid treating sparse self-disclosures as representative demographic distributions.

7 Limitations and Ethical Statement

All findings are derived at the post level within a single Reddit community. The unit of analysis is the post rather than the individual, and a single user may contribute multiple posts. Consequently, reported "prevalence" reflects post-level rates within this corpus rather than individual-level clinical prevalence. In addition, the dataset is restricted to a specific platform, time period (2008–2021), and self-reported content, which may limit generalizability to other populations or platforms.

The keyword-based filtering strategy, while interpretable and reproducible, inherently favors explicit lexical expressions and may miss implicit, metaphorical, or emerging terminology. Although validation confirms high precision, recall was not estimated due to the absence of full-corpus annotation. As a result, some relevant posts may be missed, and the reported co-occurrence rate should be interpreted as a lower bound. More context-sensitive approaches may address these limitations in future work.

Finally, comorbidity and substance use patterns were not compared against non-ED posts, limiting the ability to determine whether observed patterns are specific to ED-related posts or reflect broader trends in suicidal discourse. In addition, demographic analyses rely on small subsets of posts with self-disclosed age or gender and should be considered exploratory rather than generalizable.

Ethical Statement: The data were originally sourced from a publicly available Kaggle dataset. No identifiable user information was collected, used, or disclosed in this analysis. Because the posts concern sensitive mental health experiences, findings are reported only at the aggregate post level, and the results should not be used to infer individual diagnoses or clinical risk.

8 Summary and Conclusion

This study provides a corpus-level characterization of explicit eating-disorder references in suicidal-distress posts from Reddit's *r/SuicideWatch*. Among 27,434 posts, 688 contained ED-related content, corresponding to a post-level co-occurrence rate of 2.5%. The most frequent ED-related indicators included broad self-identification terms such as *eating disorder*, as well as specific diagnostic, behavioral, and perceptual terms such as *anorexia*, *calories*, *binge eating*, *bulimia*, *body image*, and *body dysmorphia*.

Within the subset of posts containing self-disclosed demographic information, adolescents, young adults, and gender-minority individuals were represented, although these findings are exploratory because demographic information was sparse. A substantial portion of ED-related posts referenced co-occurring mental health conditions, most commonly depression, anxiety, trauma/PTSD, and guilt/shame. Substance-use references appeared in 27.0% of posts, with alcohol-related terms appearing most frequently.

The findings suggest that ED-related suicidal-distress discourse often contains overlapping psychological and behavioral references. For NLP research, this supports transparent corpus construction, careful interpretation of post-level labels, and modeling approaches that can represent multiple co-occurring textual signals. Future work could compare ED-related posts against non-ED suicidal-distress posts, annotate a larger portion of the source corpus to

estimate recall, refine keyword coverage, and examine more context-sensitive approaches to identifying ED-related language in online mental health discourse.

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